

Employment Status and Credit Cycles: An Extensive Margin Analysis in Heterogeneous Agent Life Cycle Models.

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Research Question & Motivation

Motivation:

- The Global Financial Crisis demonstrated income loss and deleveraging occur together: unemployment surged while household debt fell sharply.
 - Greece: +20.5pp unemployment vs. −30.9% household debt
 - Spain: +18.3pp unemployment vs. −32.4% household debt
 - Portugal: +9.2pp unemployment vs. −27.4% household debt
 - Italy: +6.9pp unemployment vs. −8.3% household debt
- Standard models miss this: RBC/NK use representative agents, and HANK models adjust labor on the intensive margin, ruling out endogenous tightening of employment based borrowing constraints.
- Existing extensive margin models omit intergenerational lending and employment based borrowing constraints.

Research Question & Motivation

Research Question:

- How does introducing employment risk, switching from being employed with credit access to being unemployed and borrowing constrained, that changes over the life cycle, affect household borrowing, saving, interest rates, and utility?

Addressing the Gap

Captures Unemployment's "Double Shock":

- Unemployed lose income and access to credit because borrowing is tied to employment status.

Intergenerational Lending (OLG):

- Young borrow from middle aged savers, matching real world dynamics

Labor Adjusts on Both Margins:

- Firms decide effective labor levels that fail to align with the utility maximizing preferences of the household
- Extensive (headcount) vs. intensive (hours) margins compared.

What This Allows Us To See:

- How labor dynamics drive saving, consumption, the loanable funds market, and utility.
- Policy analysis of how counter cyclical transfer policies affect the model.

Extensive Margin in Business Cycle Models

RBC: Hansen (1985)

- Indivisible labor lets hours adjust along the extensive margin, so total hours are more volatile than productivity.
- Variance of total hours ($T = E \times H$): **55%** employment, **20%** hours per worker, **25%** covariance.

Variables are percent deviations from trend (quarterly HP filter, $\lambda = 1600$).

NK: Galí, Smets, and Wouters (2012)

- Real wage rigidity amplifies labor market volatility and helps match unemployment data.
- Unemployment as an observable gives the “correct” output gap.
- Limitation: the model is built for labor market analysis, and full risk sharing ($C_t(i, j) = C_t$) is assumed across labor types and disutilities of work.

Continuous Time Heterogenous Agents

Microeconomic Heterogeneity and Macroeconomic Shocks, Kaplan and Violante (2018):

- Examine how adding household heterogeneity caused by idiosyncratic productivity shocks influences the macroeconomic response to aggregate shocks.
- They looked at Representative Agent and Heterogeneous Agent models, up to the two-asset HANK.
- “The fact that labor market risk is mostly exogenous in current HANK models is a gaping hole in the framework.”

Heterogeneous Agent Papers With the Extensive Margin

Existing literature: Krueger et al. (2016), Rendahl (2016), Challe et al. (2017), McKay and Reis (2016), Ravn and Sterk (2017) show that unemployment risk shapes saving, consumption, and the effectiveness of fiscal and insurance policy. **What isn't included:**

- **Continuous time:** models are in discrete time, so continuous employment transitions are not captured.
- **Flexible asset positions:** restrictive borrowing limits force agents to hold zero assets, omitting how past borrowing or saving affects current decisions.
- **Employment based borrowing constraints:** without them, the double income shock (job loss tightening credit) is missing.
- **Life cycle framework:** decisions that depend on life stage and past employment paths are absent. An OLG structure would also allow natural intergenerational borrowing and lending.

OLG Literature and Labor Market Flow Dynamics

OLG literature:

- **Eggertsson, Mehrotra, & Robbins (2019):** an OLG endowment economy of secular stagnation.
- **Kaplan, Mitman, & Violante (2020):** an OLG model in which multiple shocks drive house price fluctuations, with debt subject to a maximum payment to income ratio and a minimum payment that forces amortization before retirement.

Labor market flow dynamics:

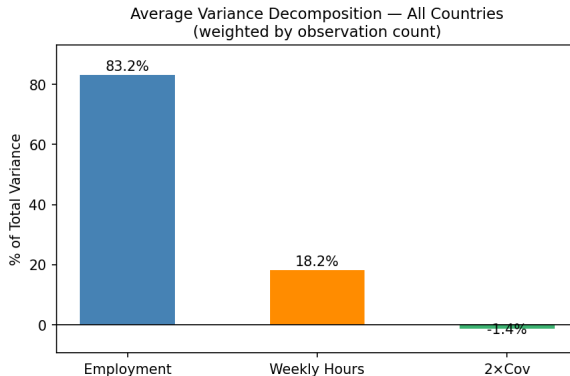
- **Shimer (2012):** in a continuous time framework, movements in the job finding probability account for most fluctuations in unemployment (77% since 1948, 90% after 1987).

Data used

European Data As a Complement:

- Data is Quarterly from 1996 to 2019
- Greater cross country variation that can be exploited for identification.
- The US lacks comprehensive state level household debt data, limiting the scope of within state debt analysis.
- European empirical evidence provides additional clarity by capturing a broader set of shocks and institutional differences that a single country study may miss.
- Specifically, Europe's demographic cycle is 10 to 20 years ahead of the US. Meanwhile, some European countries (Portugal, Italy, Greece, and Spain) have faced public debt issues, a challenge the US is becoming increasingly likely to face as its debt grows.

Variance Decomposition



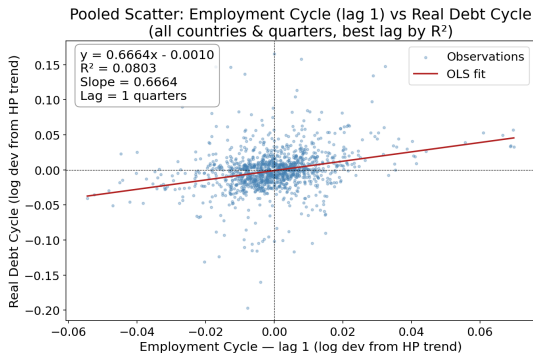
- Employment dominates total variance.
- correlation is barely negative because of Austria's negative covariance.

Variance Decomposition Within Country

Variance Decomposition of Total Hours — All Countries



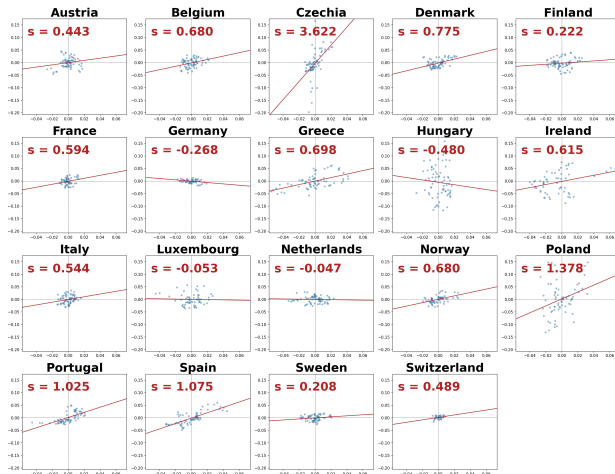
Employment vs Debt Aggregate



- There is a strong correlation between real detrended household debt and detrended unemployment lagged by one quarter/period.
- Unemployment was lagged due to its empirical significance.
- Detrending method follows the same framework as Hansen 1985

Employment vs Debt Country Specific

Employment Cycle (lag 1) vs Real Debt Cycle — By Country



OLG Overview

Key Details

- The young agents will be employed or unemployed.
- The model is used to highlight how loan demand responds when unemployment variation is restricted to youth.

OLG Dynamics

Key Details:

	Income	Borrowing/Dissaving	Consumption
Young Unemployed	+	0	+
Young Employed	++	+	+++
Middle Age Employed	+++++	- -	+++
Old Retired	++	+	+++

Borrowing

- There is a lifecycle consumption smoothing dynamic
- Borrowing constraint is based on income.
- Borrowing constraint is an exterior solution.

OLG Setup

- **Utility Function**

$$U_t^y = \ln(C_t^y) + \beta \ln(C_{t+1}^m) + \beta^2 \ln(C_{t+2}^o)$$

- **Debt Constraint**

$$D_t = B_t(1 + r_t) = \gamma Y_t^y$$

- **Euler Equation**

$$\frac{1}{C_t^m} = \beta E_t(1 + r_t) \frac{1}{C_{t+1}^o}$$

OLG Setup cont.

• Loan Demand

$$L^d = \frac{(1 - \omega_t)D^e + \omega_t D^u}{1 + r_t} = \frac{(1 - \omega_t)D^e}{1 + r_t}$$

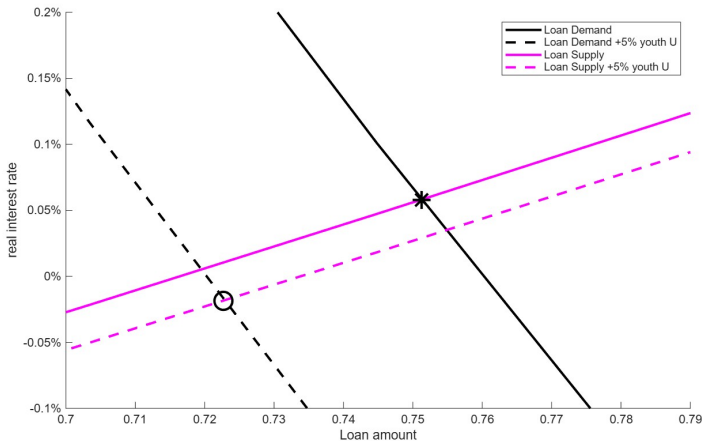
• Loan Supply

$$L^s = \left(\frac{\beta}{1 + \beta} \right) (Y_t^m - (1 - \omega_{t-1})D^e) - \left(\frac{1}{1 + \beta} \right) \frac{Y_{t+1}^o}{1 + r_t}$$

• Interest Rate

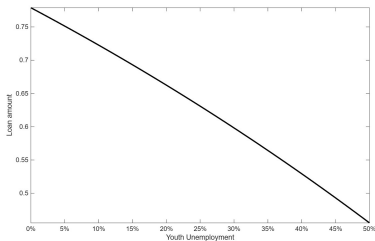
$$1 + r_t = \frac{(1 + \beta)(1 - \omega_t)D^e + Y_{t+1}^o}{\beta(Y_t^m - (1 - \omega_{t-1})D^e)}$$

Results: Youth Unemployment's Comparative Static

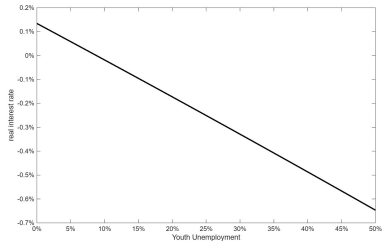


Notes: Highlights the effect of youth unemployment on loan demand and loan supply.

Results: Youth Unemployment on Total Loans and Interest Rate

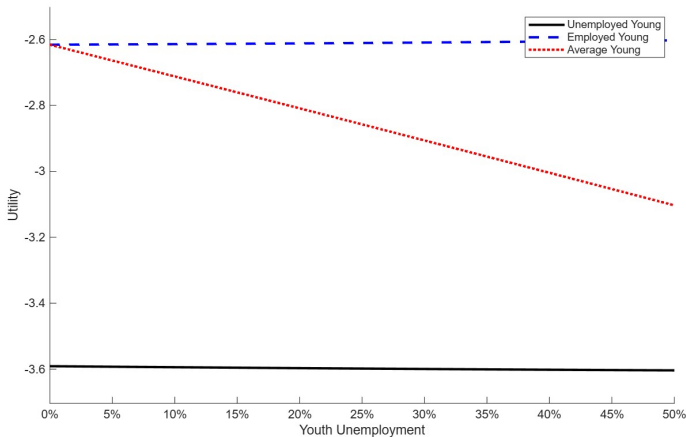


Notes: The loan amount drops at higher employment rates.



Notes: Interest rates drop as youth unemployment goes up.

Results: Youth Utility



Notes: Average youth utility drops when youth unemployment increases.

HA Overview

Goal of HA model

- Move beyond OLG toward a more empirically grounded heterogeneous agent lifecycle model

Key Additions:

- General equilibrium model with a firm setting the effective labor level based on productivity level.
- Life cycle income dynamics.
- Employment transition dynamics based on Shimer (2012).

HA Setup

Key Details:

- Discrete 3 axis grid with values based on asset level, employment status (employed/unemployed), and age (20 - 70 w/ death at 75).
- Income is set by age and employment status
- Unemployment/Social Security is .33x of the average wage income in equilibrium.
- The unemployed cannot borrow, but must maintain existing payments out of their unemployment payments.
- There is a precautionary savings motive to hold positive assets. This allows agents to smooth consumption when they are unemployed.

HA Setup continued

• Value Function

$$\rho v(a, e, l) = \max_c u(c) + v'_a(a)\dot{a} + \sum_{e' \neq e} \lambda_{ee'} [v(a, e', l) - v(a, e, l)] \\ + \sum_{l' \neq l} \lambda_{ll'} [v(a, e, l') - v(a, e, l)]$$

• Budget Constraint

$$c(a, e, l) = (wz_l h_e + u_e(1 - e) + \pi_l)(1 - \tau) + ra - s(a, e, l)$$

• Kolmogorov Forward Equation

$$\partial_t g_t(a, e, l) = \partial_a [s(a, e, l) g_t(a, e, l)] - g_t(a, e, l) \sum_{e' \in E} \lambda_{ee'} + \sum_{x \in E} \lambda_{xe} g_t(a, x, l) \\ - g_t(a, e, l) \sum_{l' \in L} \lambda_{ll'} + \sum_{x \in L} \lambda_{xl} g_t(a, e, x) t$$

Market Clearing Conditions

- **Asset Market Clearing Condition**

$$A(r) = 0 = \sum_e \int_l \int_a ag(a, e, l; r) da$$

- **Government Budget Clearing Condition**

$$\begin{aligned} gov_surplus = 0 = & \sum_e \int_l \int_a \tau_1(w(z_l h_{e,t}) + \pi_l) \\ & - \sum_e \int_l \int_a u_e(1 - e) \end{aligned}$$

Savings Motive

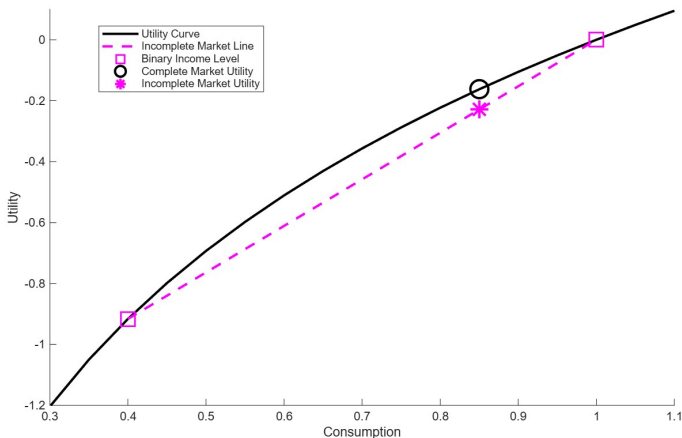
Three Savings Motives

- Intertemporal Savings Motive: savings concerning r and ρ .
- Smooth Consumption: smooth savings over the life cycle.
- Precautionary motive: occurs when the convexity of the marginal utility $u''' > 0$.

Effect of Unemployment Increase

- The borrowing constrained unemployed agents cannot borrow to smooth consumption.
- The motivation for precautionary saving is higher among individuals with low initial assets when the risk of unemployment becomes more likely with a longer unemployment duration.

Incomplete Markets



Notes: There is a motivation to self insure when facing the same expected income when there is a range of possible incomes.

Asset Movement

- The amount of debt that can be taken out (negative asset level) is based on a payment to income ratio and a minimum debt payment that amortizes at retirement age.
- Payment to income PTI based on Quercia, R. G., McCarthy, G. W., & Wachter, S. M. (2003) value of .28.

$$pmt_t = .28y_{e,t}e_e$$

- Minimum Payment amount

$$pmt_t = \frac{-a * r}{1 - (1 + r)^{65-t}}$$

- Assets at the time of death will be given to the new group of 20-year-olds as a bequest from dying grandparents to grandchildren.

Life Cycle Movement

- The income is based on the income by age in the 2013 report by the BLS. The data was fit with a parabola for simplicity.
- The dividend is based on the same methodology. All employed individuals get the same dividend income by age.
- There is forced retirement at age 65. Everyone provides a perfectly inelastic labor supply when below age 65.

Firm Dynamics

- Effective labor used is the sum of on the productivity by age z_l times hours worked by all agents $h_{e,t}$.

$$N_t = \sum_e \int_l (z_l h_{e,t})$$

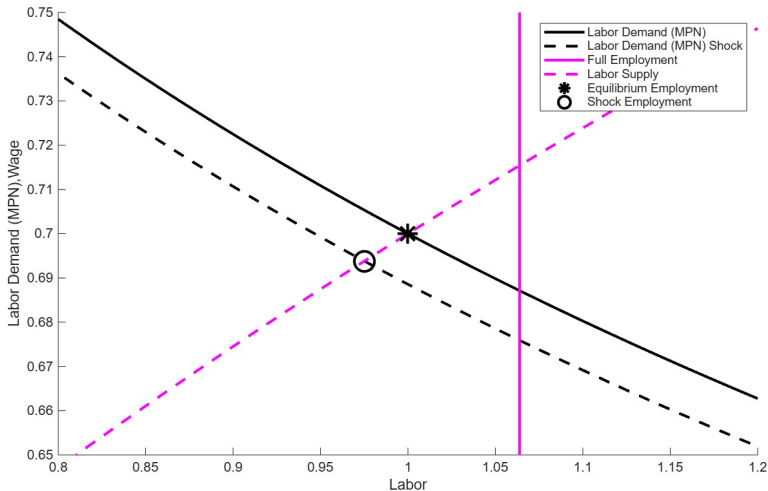
- Standard profit function exists

$$\pi_t = A_t(N_t)^{1-\alpha} - w_t N_t$$

- Labor demand is determined by

$$N_t = \left(\frac{A_t(1-\alpha)}{w} \right)^{\frac{1}{\alpha}} = A_t^{\frac{1}{\phi+\alpha}}$$

Labor Market With Shock



Notes: Downward wage rigidity and a negative productivity shock drive unemployment. This is a divergence from when agents choose the hours they work.

Employment Movement

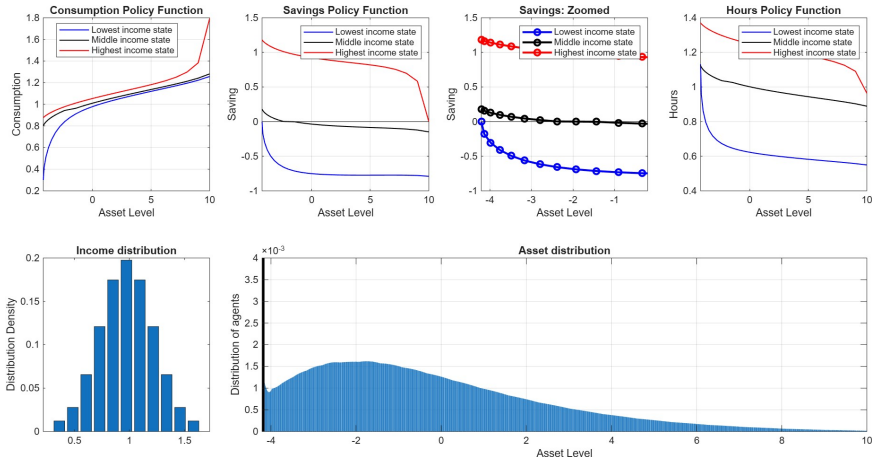
Employment Flows (Shimer 2012)

- From 1948 to 2010, the quarterly average of monthly hiring rates is 44 %. This corresponds to a 82.4 % quarterly hire rate.

Unemployment Dynamics used to enrich the model

- The unemployment of 20-24 year olds is 1.7x total unemployment.
- 25-34 unemployment roughly matches the total unemployment rate.
- The separation rate is set to ensure that total unemployment matches the natural rate.
- The chance that 20 and 25 year olds exit employment is set to be higher.
- A portion of 20 year olds will start employment by taking some of the jobs vacated from the retiring 65 year olds.

Steady State Dashboard: Modified Huggett Model



Notes: Asset distribution suffers from the missing middle. Agents here choose how much they work.

Intensive vs Extensive Margin vs Huggett Model

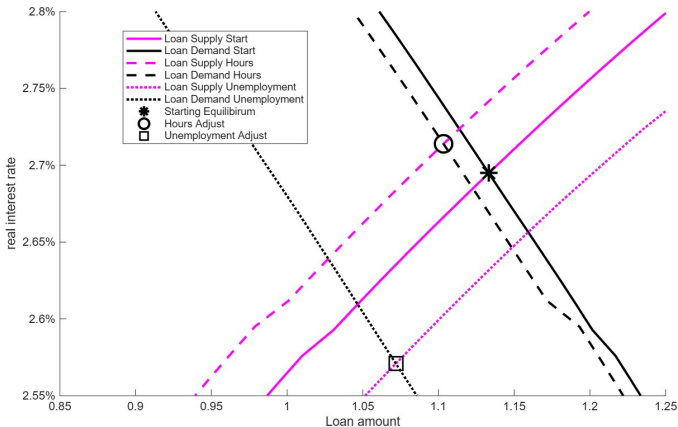
Intensive vs Extensive Margin

- After a productivity shock, total hours worked drop.
- The two labor margins (extensive vs intensive) will be compared.
- This allows us to highlight differences between the two margins.

Modified Huggett Model

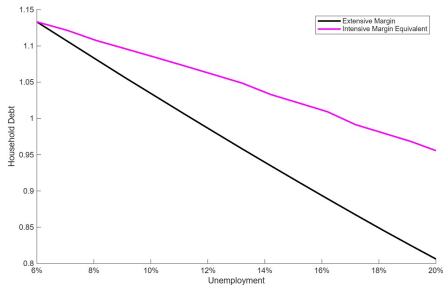
- The modified Huggett model is used to demonstrate that it fails to capture similar dynamics in the loanable funds market.
- Huggett model doesn't have significant changes in the precautionary savings motive.

Results: HA Comparative Static Loanable funds market

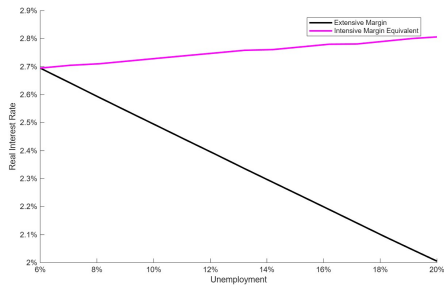


Notes: The extended margin has large effects on both the loan supply and loan demand.

Results: Comparative Static Loan Quantity and Interest Rate

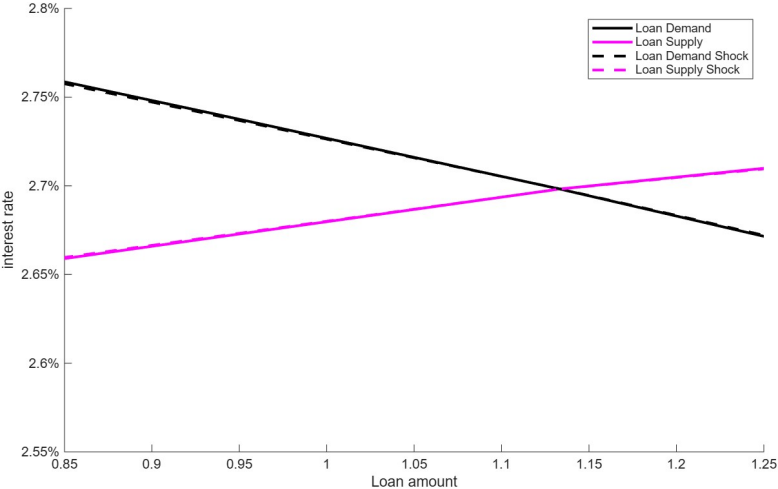


Notes: The loan amount drops at faster over the extensive margin.



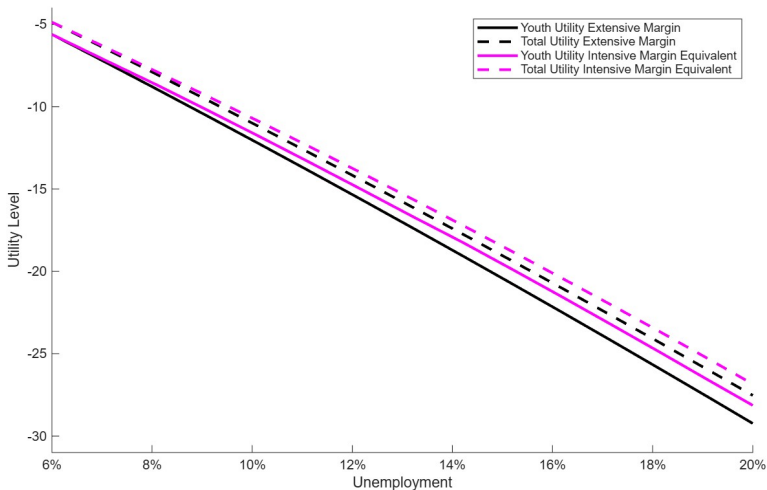
Notes: Interest rates drop over the extensive margin and rise in over the intensive margin.

Results: HA Comparative Static Modified Huggett



Notes: Little to no effect in the comparative static for the Huggett model.

Results: Utility HA model



Notes: The extensive margin has a larger utility drop

Introducing Countercyclical Transfer Payments

Benefit of adding Countercyclical Transfer Payments

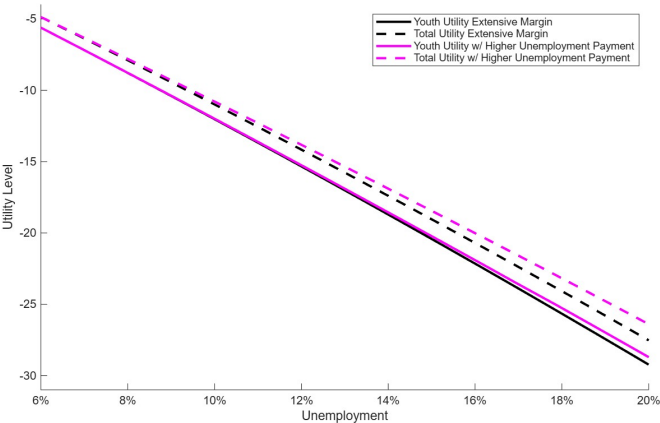
- Adverse productivity shocks lower output, income, and utility.
- Raising social security and unemployment transfers mitigates the decline in household utility (vs. the fixed transfer payment benchmark).

What The Payments Are

- Transfers are inversely proportional to output:

$$T_{0,shock} = T_{0,ss} \times \frac{Y_{ss}}{Y_{shock}}.$$

Utility Increase Transfer Payments



Notes: Increasing transfer payments increases utility.

HA IRF Overview

Benefits of IRF

- IRF allows us to compare the IRF to both a temporary and a permanent productivity shock in the intensive margin and extensive margin models.
- This will highlight the impact of incorporating employment risk on the saving and borrowing dynamics.

Variables of Interest:

- Real interest rates, household loans outstanding, percent of individuals borrowing constrained, hiring transition probabilities, and the unemployment rate.

IRF Methodology

Overall Methodology

- I use the Achdou, Han, Lasry, Lions, and Moll (2022) continuous time approach to solve the model.
- The value function (HJB) is solved backward, and then the distribution is solved forward (KFE).

Labor Adjustment Cost

- This is added because transitioning employment instantaneously doesn't make sense.
- Lechthaler & Snower (2013) added quadratic labor adjustment costs to a new Keynesian model.

Solving the IRF

HJB and KFE

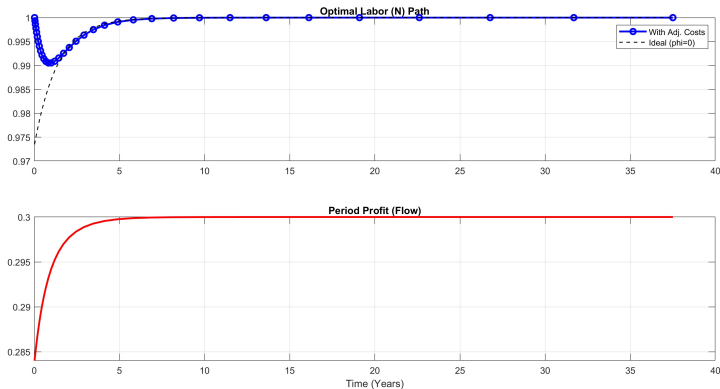
- HJB is solved backward, and KFE is solved forward.
- Three residuals: labor market, government budget, and asset residual are solved in order.

Effective Labor Determined by:

$$V = \max_{N_t} \int_0^{\infty} e^{-\rho t} (A_t(N_t)^{1-\alpha} - wN_t - \frac{\theta}{2} \dot{N}_t^2) dt \quad (1)$$

- Labor selected maximizes the firm's profit function.
- Adjust the hiring rate in the Extensive Model.
- Adjust hours worked by agent in the Intensive Model (N_t).

Labor IRF



Notes: Adding adjustment costs dampens and delays the labor response.

Solving the IRF cont.

Taxes

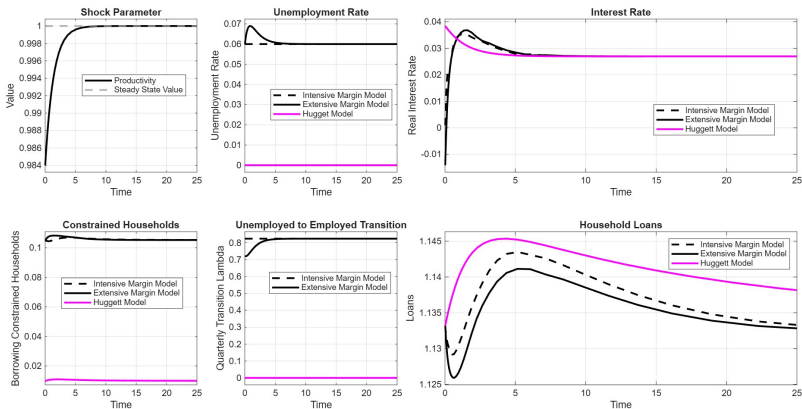
- Taxes depend on the wage income, dividend income, and the number of unemployed agents. This can be solved directly after the labor market.

Interest Rate

- Interest rates have no effect on the labor market and taxes, so they can be solved independently after the labor market and government budget.
- Clear asset market $A(t) = 0$ at all time
- A smoothing function is used to smooth the interest rate path.

Temporary Negative Productivity Shock

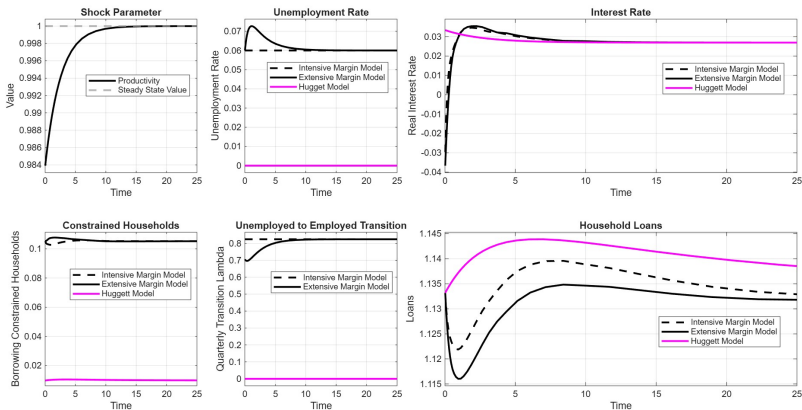
IRF Combined Dashboard



Notes: Large standouts are the interest rate paths with the Extensive Margin Model having the lowest bottom before both spike above the Huggett Model. Both the Extensive and Intensive Models have an initial contraction in household debt before increasing.

Higher Persistence Shock Productivity Shock

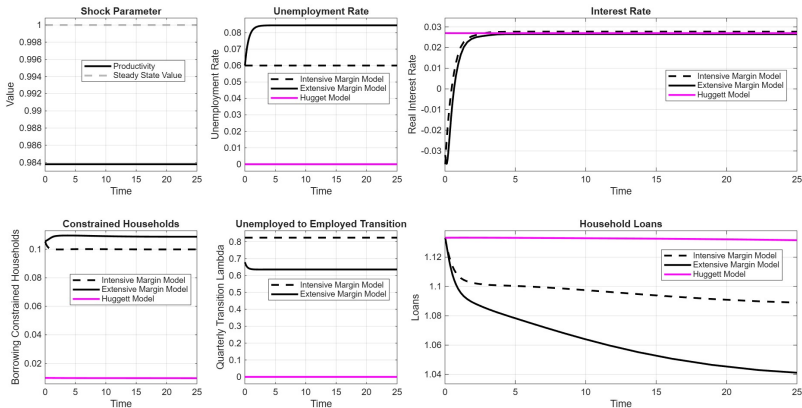
IRF Combined Dashboard



Notes: Increasing shock persistence to .9 creates the dynamics where the extensive margin does not overshoot the starting household loan value.

Permanent Negative Productivity Shock

IRF Combined Dashboard



Notes: Difference in leverage response is large and interest rates have persistent differences.

Empirical Overview

- Vector autoregression (VAR) and Local Projections (LP) estimate the effect of a productivity shock on real-world data.
- Quarterly variables: labor productivity, weekly hours, unemployment rate, and real household debt.
- Weekly hours and unemployment capture employment dynamics.
- Real household debt is the most stable series to compare with the theoretical models, because interest rates are heavily affected by default probabilities.

Data Overview

- **Sample:** quarterly data, 1998 to 2019 (earliest date common to all sources, ending just before Covid).
- **Eurostat:**
 - Labor productivity (used instead of multi-factor productivity, which is only annual)
 - Harmonized consumer prices
 - Average weekly hours worked
- **OECD:** unemployment rate
- **BIS:** household debt
- **Resulting panel:** labor productivity, weekly hours, unemployment, and real household debt across European economies, used to study employment dynamics and household debt after productivity shocks.

VAR Structure

- The VAR was calculated with a Cholesky decomposition of errors.
- The order of the Upper triangle was as follows: labor productivity, weekly hours, unemployment rate, and household debt.
- The VAR imposes an AR(1) on the productivity shock, and all other variables are lagged by 4 quarters.

VAR Formula

VAR formula:

$$(I - \Phi_1 L - \Phi_2 L^2 - \Phi_3 L^3 - \Phi_4 L^4) X_t = e_t \quad (2)$$

This is a reduced from VAR that forces labor productivity to be an AR(1)

$$\Phi_1 \text{ layout} \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \quad \Phi_{2,3,4} \text{ layout} \begin{bmatrix} 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$X_t = \begin{bmatrix} A_t : \text{LaborProductivity} \\ H_t : \text{WeeklyHoursWorked} \\ U_t : \text{UnemploymentRate} \\ B_t : \text{RealHouseholdDebt} \end{bmatrix}$$

VAR Error Terms

Cholesky Error Terms:

$$e_t = Bu_t$$

Matrix Form:

$$\begin{bmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \\ e_{4t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ b_{21} & 1 & 0 & 0 \\ b_{31} & b_{32} & 1 & 0 \\ b_{41} & b_{42} & b_{43} & 1 \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \\ u_{4t} \end{bmatrix}$$

Local Projection

Local Projection:

- Impulse response functions by running a sequence of OLS regressions for different horizons (h).

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma_h x_t + v_{t+h}$$

α_h constant, β_h Reaction to shock, and $\gamma_h x_t$ control variables

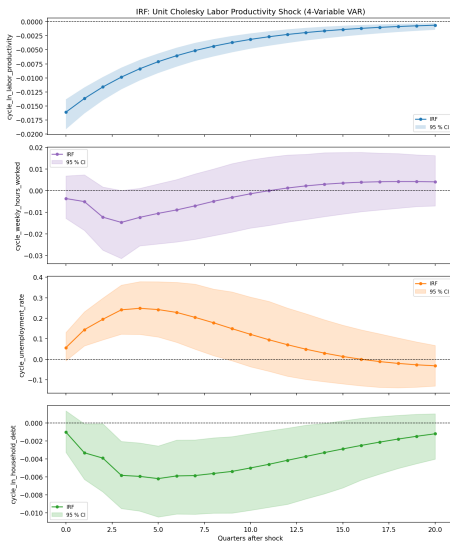
- Residual MA(h) Error Term:

$$v_{t+h} = u_{t+h} + \psi_1 u_{t+h-1} + \dots + \psi_h u_t$$

VAR

VAR shows the following:

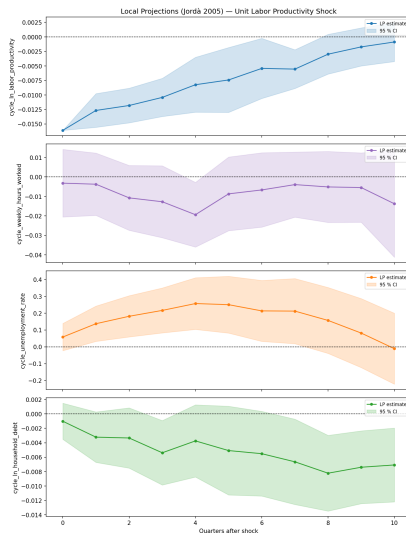
- Ln detrended productivity follows an AR(1) structure and shows the difference in log levels.
- Weekly hours worked don't move significantly.
- The UR increases significantly.
- Ln detrended real debt decreases significantly.



LP

LP broadly follows the VAR findings:

- Ln detrended productivity now moves freely.
- Weekly hours has minimal movement.
- The UR increases significantly.
- Ln detrended real debt decreases significantly over time.



Split Countries to Look at Difference

Why countries are split

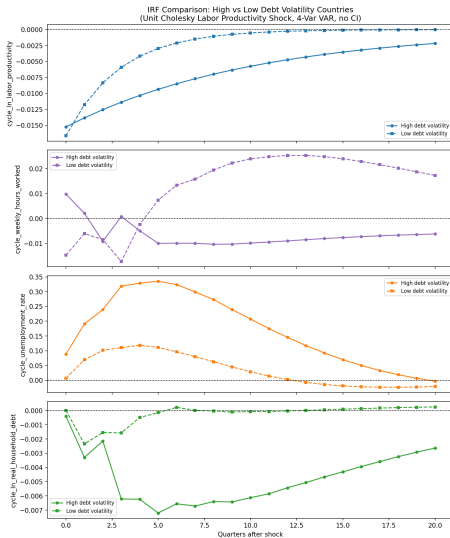
- Countries are split to see if there are significant differences in the IRF based on the splits

Countries Split by the following Groups

- Household Debt Variance
- Household Debt to GDP

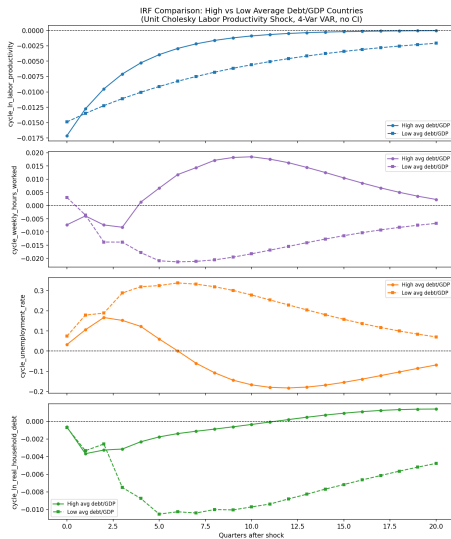
VAR Household Debt Volatility

- The countries are split based on the variance in cyclical real debt.
- The median variance 0.003472.
- High debt volatility countries (8): Czechia (var = 0.023705), Finland (var = 0.005124), Greece (var = 0.009867), Hungary (var = 0.019311), Italy (var = 0.004194), Poland (var = 0.012563), Portugal (var = 0.009483), and Spain (var = 0.010687)
- Low debt volatility countries (8): Austria (var = 0.001450), Belgium (var = 0.002750), Denmark (var = 0.002155), France (var = 0.000897), Germany (var = 0.000374), Ireland (var = 0.001910), Norway (var = 0.000353), and Sweden (var = 0.000665).



VAR High vs Low Household Debt/GDP

- The countries are split by on mean debt to GDP ratios.
- Median loans-to-GDP: 51.0
- High average household debt/GDP countries (8): Denmark (mean = 107.8), France (mean = 51.6), Germany (mean = 56.9), Ireland (mean = 78.6), Norway (mean = 78.5), Portugal (mean = 74.3), Spain (mean = 65.6), and Sweden (mean = 71.6).
- Low average debt/GDP countries (8): Austria (mean = 50.1), Belgium (mean = 49.8), Czechia (mean = 21.5), Finland (mean = 50.5), Greece (mean = 45.9), Hungary (mean = 22.6), Italy (mean = 34.9), and Poland (mean = 26.7).



Empirical Findings

- The empirical responses to productivity shocks show that the movement in the unemployment rate is significant, and weekly hours are not. They also show that real household loans are significantly affected by productivity shocks.
- Countries with high cyclical real debt variation and lower average household debt to GDP have similar productivity shocks, but the persistence is longer.
- This implies that countries experiencing persistent shocks experience larger increases in unemployment and, therefore, larger decreases in household loan balances.

Conclusion

- This model captures the "double shock" of unemployment in a life cycle model designed to assess its implications.
- The extensive margin model highlights a drop in real interest rates, loan quantities, and utility following a negative productivity shock.
- These findings are consistent with the empirical findings that the VAR and LP generated.

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Lack of ability to spend and borrow from external sources

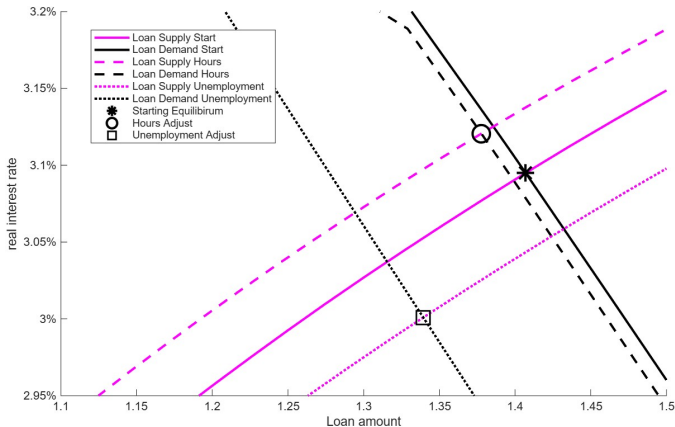
Forbes Article: 63% Of Americans Don't Have Enough Savings To Cover A \$500 Emergency

- This model simplifies all assets into one account, where in reality, individuals might have loans and a positive savings account.
- This is limited because most people don't have that much money in an extra savings account.

Liberty Street Economics: 18 percent of borrowers were using at least 90 percent of their available credit

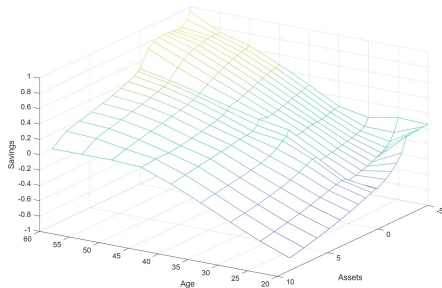
- Credit cards allow people to smooth consumption even if they are in debt.
- This is limited because a good number of individuals are already at or near their credit card limit.

HA Comparative Static with G and B_{gov}

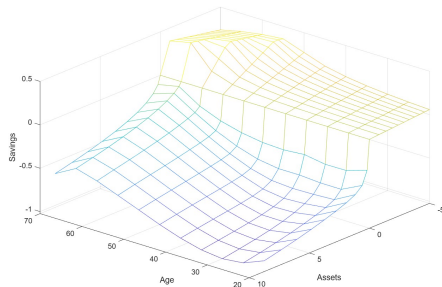


Notes: With G is set to .17 and B_{gov} is set to 70% annual GDP, the movement in supply and demand are reiterated

HA Saving Curves

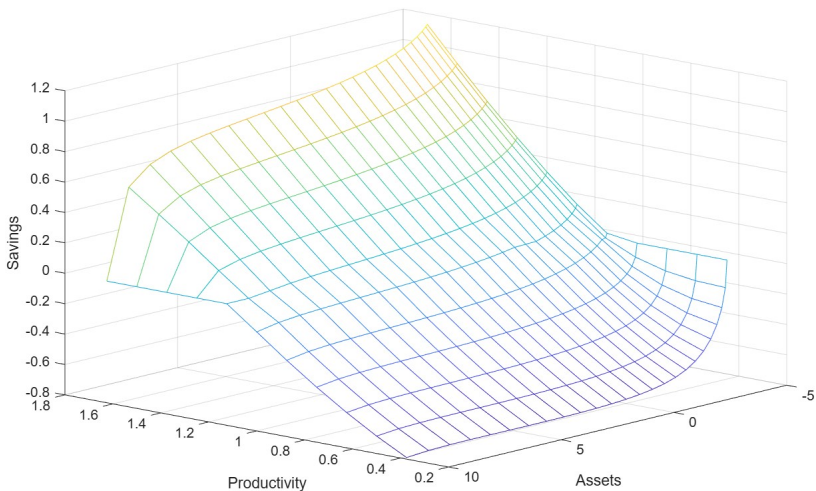


Notes: Saving is higher with age and with lower asset levels.



Notes: Saving is negative above the zero asset borrowing constraint.

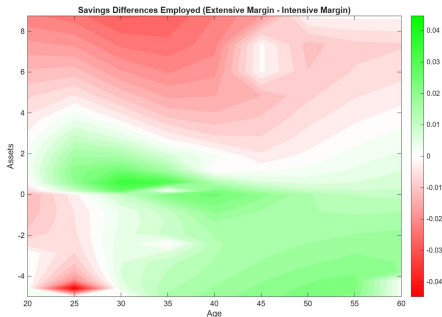
Modified Huggett Saving Curve



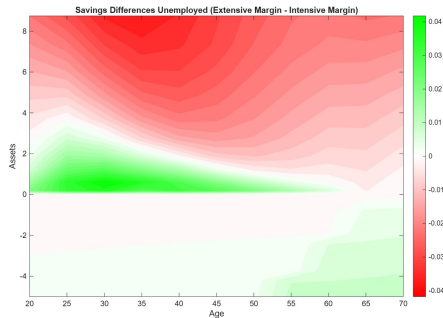
Much less intricacies than HA model.

Notes:

Savings Differential Across the Margins

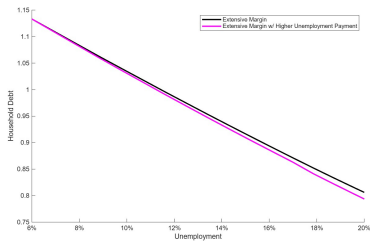


Notes: Borrowing is higher for the extensive margin early on, then less as they age.

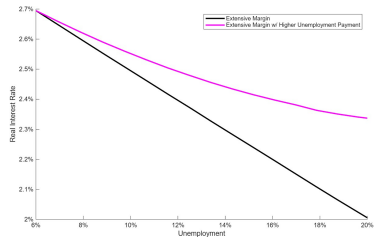


Notes: Precautionary savings increases above zero assets.

Comparative Statics Loanable Funds with Transfers



Notes: Counter cyclical transfers decrease the quantity of loans.

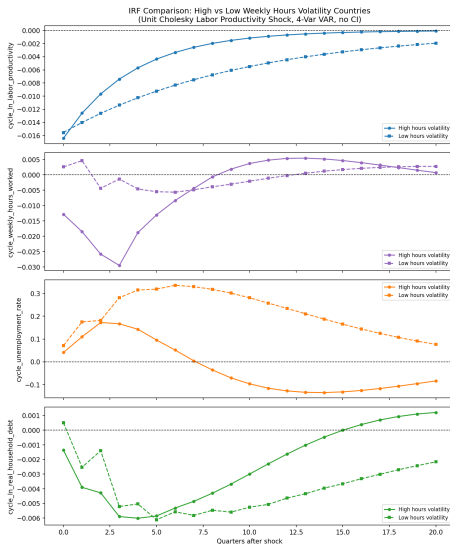


Notes: Real interest rates stop declining.

Return

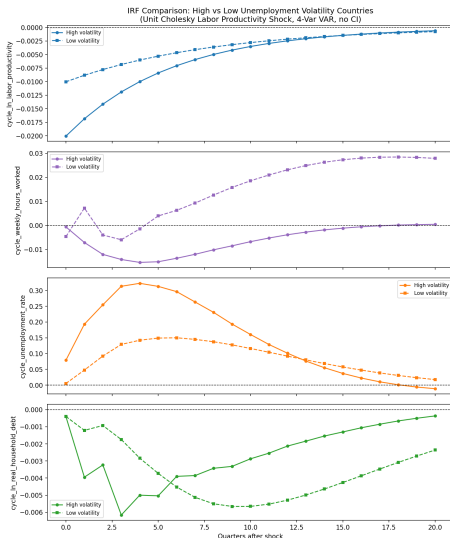
VAR Weekly Hours Variance

- The countries are split based on variance in weekly hours worked.
- The median variance: 0.061210.
- High hours volatility countries (8): Austria (0.750629), Czechia (0.132484), Denmark (0.156413), France (0.093829), Ireland (0.180871), Italy (0.168787), Portugal (0.061698), Spain (0.061316).
- Low hours volatility countries (8): Belgium (0.061104), Finland (0.016763), Germany (0.030801), Greece (0.056951), Hungary (0.043921), Norway (0.023075), Poland (0.036172), Sweden (0.014668).



VAR Unemployment Variance

- The countries are split based on the variance in the UR.
- Median variance: 2.1049.
- High UR volatility countries (8):
Czechia (var = 3.6749), Greece (var = 41.3019), Hungary (var = 2.6149), Ireland (var = 7.0445), Italy (var = 2.1902), Poland (var = 5.4764), Portugal (var = 9.2806), and Spain (var = 24.5391).
- Low UR volatility countries (8):
Austria (var = 0.3378), Belgium (var = 0.7696), Denmark (var = 2.0195), Finland (var = 0.7927), France (var = 0.3304), Germany (var = 0.0593), Norway (var = 0.3614), and Sweden (var = 0.7271).



Interest Rate Optimization

Levenberg-Marquardt

$$F_{aug} = [F; \text{sqrt}(\lambda_s) * Ps]; \quad (3)$$

$$J_{aug} = [J; \text{sqrt}(\lambda_s) * R]; \quad (4)$$

$$H = J'_{aug} * J_{aug} + \mu * \text{eye}(nT); \quad (5)$$

$$g = J'_{aug} * F_{aug}; \quad (6)$$

$$dx = -H^{-1} * G; \quad (7)$$